

Weisst Du wieviel Sternlein stehen? – Building older adults’ confidence in technology use through co-designing digital storytelling

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CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Field studies.**

KEYWORDS

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1 ABSTRACT

To help older adults learn about meaningful use of technology, we conducted a co-design digital storytelling project with a group of residents from independent living units in a residential care setting. In this case study, we present in detail the development of one participant’s story in six stages and an audio section. We reveal the challenges experienced and how they were tackled through the development of the story. The lessons we learned from this case study are 1) how the co-design process helped us explore together Christel’s emotional goals that she wanted to express in this story, and 2) how the digital storytelling allowed us to present these emotions with different materials and digital means to accurately express a precious memory. Our research contributes to how to help older adults explore and understand the possibilities of technology use, conduct meaningful practices and sustain their use in daily life.

2 INTRODUCTION

Technology provides potential opportunities for older adults to improve their quality of life from different perspectives. For instance, technology can enable access to more information and reduce social isolation [9]. Previous studies explored older adults’ attitudes towards technology and show that they tend to use technology when they perceive its usefulness [10] and when they find a purpose to use it in their everyday practices [1]. However, some works reported that although some older adults had a strong interest in learning to use technology, they were still hesitant and sometimes lacked the confidence to interact with it [2, 4, 7].

This work is part of a larger research project with the objective to help older adults to learn to use technology through meaningful activities. The case study we report is part of a co-design group activity in a local residential care setting, named “Telling our story”. The group decided to produce digital stories that captured important memories that they wanted to share with others. The aim of this activity is to use digital storytelling as a way for participants to tell stories about themselves [5] in a process where they co-produce the digital materials [12] and use technologies in ways that align with their interests and motivations.

The activity brings opportunities for participants to learn about the use of technologies and at the same time to create a meaningful outcome. During the first group sessions, we discussed the motivations and goals of the participants.

Digital storytelling incorporates technology which consists of various multimedia elements compared to oral storytelling [3]. Story creators can grow their technology skills through building the story [6, 11]. For older adults, digital storytelling offers opportunities to express themselves in new and different ways. Considering these benefits, this project was guided by the research question: How can we co-design digital storytelling with older adults where they can learn meaningful use of technologies?

This case study presents how we supported Christel with her digital story through six stages and the process of audio production. We found that the iterative co-design process helped us learn Christel’s different goals for conducting the digital story, particularly

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her emotional goals. We also discuss how confidence in technology use was built through meaningful and enjoyable experiences with different technologies and software. This project provided insights into how to help residents in group activities learn about technology in meaningful ways.

3 SETTING AND PARTICIPANTS

This research was conducted with participants who lived in independent living units in Martin Luther Homes, a residential care centre located in Melbourne, Australia. The residential care centre has an emphasis on German culture and provides spiritual care for its residents. We visited Martin Luther Homes weekly since October 2021. At the beginning of this research, lifestyle staff members helped us schedule the sessions and recruit residents who were interested in technology use. After the activity was set up, all sessions were conducted by the research team without assistance from staff members. After a few weeks, four residents became regular participants. Each participant created their own digital story and participated in group discussions. They shared their ideas and experiences with technology with each other. They also provided feedback and helped each other in the group. This paper focuses on the case of one of the participants, Christel, who attended all the sessions, co-designed her story with us and collaborated in writing this case study.



Figure 1: The four regular participants in the group activity and the researchers

3.1 Who is Christel?

Christel is one of the four regular participants in the group activity (Figure 1). Christel is 86 years old. She travelled to Australia from Germany with her family when she was a teenager after World War II. Now she and her husband live at the residential care centre. Christel is a skilled user of mobile technology: she has a smartphone, a tablet and a desktop computer set up at her place. She can manage basic use, such as sending and receiving emails and searching for information on the Internet. As she was interested in learning more about her devices, she joined the group activity.

4 METHODOLOGY

The aim of this study was to understand how residents can learn meaningful use of technologies through a co-design project. We recruited residents who are interested in technology use from the retirement village of Martin Luther Homes. Guided by the Participatory Action Research [8] approach, we collaborated with participants by following their interests and needs to conduct technology learning activities. Through many weeks of interactive discussions, we settled on a group project that we called “Telling our story” (more details in Section 5.1).

From the beginning of this project (October 2021) to when Christel had finished her story (September 2022), we conducted 32 sessions. All sessions were conducted face-to-face except for one that was organised online. It was important that all decisions were made with Christel and that she actively participated in the design and production of the digital materials. Based on the development of Christel’s story, we describe the co-creation process in six stages: Idea generation, Planning the digital story, Experiencing digital drawing, Creating animations, Structuring the story in PowerPoint, and Polishing the story. These stages allowed Christel to include the narrative and visual elements that added a personal touch and enriched her story. She explored and experienced different technologies so she could decide if she was interested in using them for her story. The group sessions also strengthened the connections between the participants, who shared their technology experiences and memories with other group members. Following the sixth stage, we present how the audio in the story was created. The following section describes the stages in detail and explains the design decisions made for each stage in this iterative and reflective process.

5 DEVELOPMENT OF CHRISTEL’S STORY

This section presents the development of Christel’s story, the challenges we encountered in the process and how they were addressed.

5.1 Stage 1: Idea generation

The idea for the digital story project was conceived during the first four sessions. We started by getting familiar with each other in the group and introducing the purpose of our participatory research. Then the participants discussed their goals on how they wanted to participate in the group activity. At the end of this stage, they suggested doing a shared project together. Inspired by an online story-sharing project the care provider conducted during the pandemic through Zoom¹, the participants proposed to conduct a digital storytelling project where they could learn to use technology to tell their stories to other residents and family members.

Christel had two stories from her childhood that she wanted to tell at the beginning. Although both stories were situated in difficult times in her life, her stories had humour and warmth. The first story she told was “Three slices of rye bread”:

In the last few years of the war, Mother and we three children lived in a village in the country about 100 km away from Berlin, where our father worked. As often as he could he came to see us and we children

¹<https://zoom.us/>

knew that, having saved some points on his ration cards, he would usually bring a big loaf of rye bread. On this occasion, perhaps because it was close to dinner time, anyhow, we children were hungry and asked if we could have some of this bread. “Ask your mum” said dad, and mum said “well, just one slice”, and dad took the bread knife and cut us a long crust and 2 slices – loaf lengthwise – each about 30 cm long and we shouted with joy! We looked at our mother and saw that she did not know whether to laugh or cry – she was happy to see our delight but, had she not planned to use this bread more wisely in the coming week?

That story made everyone in the group laugh. In the following session, she told another story about how she and her family were evacuated from Berlin to the South of Germany. It was a beautiful story and we all were moved when her voice was quavering at the end. The following is the final script of the story, which she named “The story of Weisst Du Wieviel Sternlein Stehen” (a German children’s song):

I was 8 years old, shortly before the end of the war in February 1945, when our family, that is, my mother and two younger brothers, had the chance to be evacuated from Berlin in one of just two trains that had been made available to take mothers and children to the south of Germany.

Our parents had prepared us well for this journey. We older children Peter and I had backpacks with spare clothing and our little brother’s one held a potty and his teddy. The station in Berlin was very crowded, and that we managed to get onto the train was thanks to family friends who took us children on their shoulders and then lifted us through the windows into the carriages.

My recollection of the journey, which took many days, are a series of pictures. There is one that, for instance, thanks to my mother’s foresight, our potty was the only one in our huge carriage and went from child to child and the other is that the long hours that we had to spend every day when we were standing somewhere without our engine. But that, miraculously, every late afternoon another engine took our carriages to a station of a nearby town. The stations, as indeed the whole towns were in complete darkness and we were helped out of our train by teenagers who escorted us then to a place where we found warmth, a meal and a bed for the night.

One evening, having again been met at the station by the wonderful teenagers, we were taken through a town which had been badly bombed. The road on which we walked had already been cleared but the houses on each side were tall and frightening ruins. It was a bitterly cold winter night and imagine our convoy of mothers and children walking slowly in darkness. We were tired and hungry but, when we looked up, we saw a sky with stars so clear that one

thought you could touch them. It was then that my little 3-year-old brother Berndel (Herman) started to sing our beloved children’s song “Weisst Du Wieviel Sternlein Stehen”. Mom and the children joined him, but not only we, the children and mothers walking next to us, then those in front and then those behind us and, after a while, the melody, like ebb and flow washed over us, we were all singing. At that moment I forgot all the difficulties and fears of the journey, we were to be looked after tonight. Everything was OK and I was just happy!

Finally, Christel decided to produce the second story in this digital storytelling project. As she thought the “Three slices of rye bread” was a sad story although it had humour, it reflected the difficulties that mothers had during war times to feed their children. In contrast, “Weisst Du Wieviel Sternlein Stehen” was easier to link to the emotion of happiness when she, her mum and her brothers sang the song.

Since the story is situated in the context of World War II, there were many moments and anecdotes that got the research team’s attention, but they created memories of emotions such as fear and uncertainty in Christel, e.g. children wearing fluorescent brooches while travelling on the train, and the inside of the train. In conversation, many of these details were dismissed because Christel wanted the story to move towards the happiness she felt at the moment when everyone was singing.

The decisions made on the plot were all made by Christel, and she remembered more details about the story while in conversation with the research team. The iterative reflection and conversations benefited the writing process of the plot and the clarity of the emotions, including the emotions that should be conveyed by the music used in the story (more details are explained in Section 5.7).

5.2 Stage 2: Planning the digital story

This stage took two sessions. After defining the story to be told, we discussed with Christel the digital materials that we needed to produce the story and how to do them. For instance, we needed a narrative of the story, an image of a bombed town that matched her memory without knowing which town it actually was, music for the song “Weisst Du Wieviel Sternlein Stehen” and a small animation for the stars.

In this project, we set the research as a loosely structured and iterative project that would follow the participants’ interests. Some participants, like Christel, were uncertain about which technologies would be used and how they would learn to use them. Similarly to other participants, Christel was looking forward to seeing a clear plan and agenda:

I prepared some stories and I am very happy to do that [the digital storytelling project]. However, it was pointed out that maybe I would not learn as much about how to apply animation and music to a story because you might do that in Swinburne and I might just “get something”, but I haven’t learnt how to do it, and my reason to come here and to respond to your suggestion was because I wanted to learn. So really we have to talk about what the path that you suggest

is! ... learning about phones and tablets is still very important for me, and with the story I'd just like to know – what will happen from now on, how much will I be involved?

To respond to Christel's questions, we clarified the objectives of this project and ensured participants had an active role in the production of the digital materials. This way, she could learn to use technology in a meaningful way for herself. This process could be different from other participants because each story was different and each participant decided the digital materials they would like to produce.

To concretise the context of this project, we started to work on Christel's story after this conversation. The third author (Diego), the fourth author (Sonja) and Christel searched for a photo of a bombed town in online public libraries. However, as Christel's story required a specific scene of darkness and people walking through the ruins bordering the street, we looked for an image that resembled these requirements and that could be easily edited with our additions to match the memory. This decision moved the project to the next stage.



Figure 2: Christel drawing on iPad and the landscape she created

5.3 Stage 3: Experiencing digital drawing

This stage took eight sessions. To help Christel create the scene of her story, the first author (Mia), who has a design background, planned to use Procreate² to draw together with Christel on the photo they selected. In the beginning, we expected Christel to draw a draft with pen and paper to tell us how she wanted the scene to be, then we would work on the digital version independently. Surprisingly, after we introduced the basic use of Procreate to Christel, she drew a landscape quickly (Figure 2) and we learned that she used to draw with watercolours which we didn't know before. She was delighted to use a tablet to draw and insisted to install a similar program on her own device so that she could practice in her own time. The interest of Christel motivated us to give her the lead in the editing process of the digital images.

In the following sessions, we introduced Sketchbook³ to Christel, as it was supported by her tablet and it is a free app. We also gave her a stylus for drawing. Christel got used to the app quickly and created the train scene for her story on her own (Figure 3). She also used Sketchbook together with her granddaughter when she visited her, and they even found some features that Mia had never

used before. During the following sessions, Christel drew a draft of the families walking through the ruins on top of the photo to help Mia create that scene. To draw the shining stars, Christel also did research online about the star constellations that can be observed in Germany.



Figure 3: The train for Christel's story

In Christel's experience of digital drawing, we were hesitant to introduce either Procreate or Sketchbook to her at the beginning, as we assumed they were professional digital drawing tools that required different interaction gestures and professional knowledge from the design domain, such as the concept of layers. We thought it would not be easy to learn for Christel because she was beginning to learn to use her tablet for drawing. However, after Christel showed great interest in the digital drawing, we decided to show more features she might need but in simple ways. We let her experience basic features first, such as selecting brushes and colours, and then introduced more tools in the program when she needed them.

5.4 Stage 4: Creating animations

This stage was conducted during six sessions. After we created the main digital materials of Christel's story, she expected to animate the train to match what happened in the story, "The long hours every day we were standing somewhere without our engine, but that, miraculously, every late afternoon another engine took our carriages to a station of a nearby town". To do this, Mia presented the concept of frames to Christel to explain how they can be used to produce an animation. Sketchbook cannot create a GIF file so we found an online website⁴ to do it. Christel created each frame by moving the train forward and Mia imported these images and uploaded them to the website to create a GIF file. As Christel wanted to create a smooth animation, the creation of the frames and the animation took significant time and effort. After creating the animation, Christel and Mia were excited and showed the final result to the group. Christel said that "That will be the highlight of the story, my grandchildren will be excited when they see that part."

In this stage, the most challenging part was not to explain the concept of frames and animations to Christel but to export the

²<https://procreate.art/>

³<https://www.sketchbook.com/>

⁴<https://ezgif.com/maker>

frames and create the GIF using websites that were not accessible and that Christel could not read properly. The interface used small fonts and there were many distracting elements, such as banners and ads. To manage that, Mia sat down with Christel during the group activity. While Christel did the work in Sketchbook, Mia did the work on the website. During the process, both Christel and Mia made mistakes but checking and fixing things together was part of the learning process. Christel was sometimes confused by the contents in different layers, when she saw Mia check which part drew at which layer in Sketchbook, Christel said, “You also need to check the layers!” showing she completely understood the concept.

5.5 Stage 5: Structuring the story in PowerPoint

After creating the animation, we spent eight sessions structuring the story in PowerPoint. Initially, we showed a YouTube video⁵ to the group about how to use Keynote to present a story with animations. Then, we explained the similarities between Keynote and PowerPoint and divided the group into those who could use PowerPoint and those who could use Keynote. Christel learned the basics of creating PowerPoint presentations with Diego.

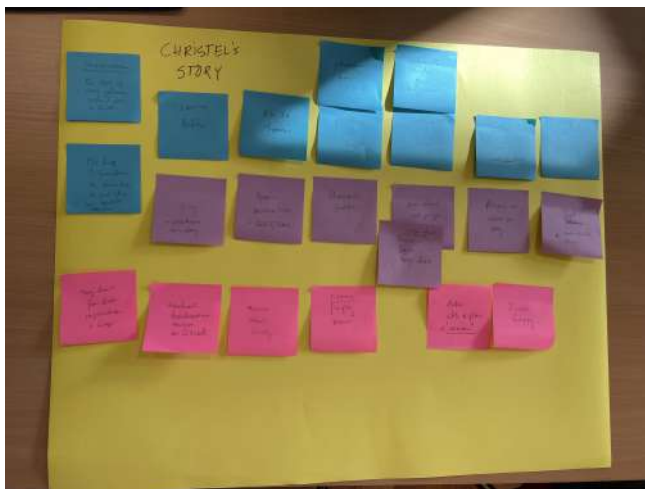


Figure 4: Christel's story mindmap

After the trial of PowerPoint, Christel wanted to present her story using PowerPoint. We helped Christel create a mindmap (Figure 4) about her story to organise her materials and record the emotions she wanted to express. Blue notes on the mindmap represented the plot in her story, purple notes represented the materials needed for the plot, and pink notes stood for the emotions she wanted to express. For instance, “star” is on a blue note, the purple note under it says “Focus on stars in sky” and the corresponding pink note contains “I was happy”, which means the view should focus on the sky when she talked about the stars and she wanted to convey the emotion of happiness to the audience. Following the mindmap, the research team put all the content into a PowerPoint file and then discussed the design of the slides with Christel in the online session. She emphasised the movement of the scene in the bombed town

⁵<https://www.youtube.com/watch?v=GTvFw3CzYUA>

should follow her narrative (detailed in 5.7) and then should focus on the shining stars at the end.

Both the trial in PowerPoint and the creation of the mindmap helped Christel understand how to present her story. After that, she could clearly see the progress of the project, how much work had been done and how much was left through organising all the materials that the story needed on the mindmap.

5.6 Stage 6: Polishing the story

After we finished the first version of the story in PowerPoint, we used four sessions to work on the final details of the story. We adjusted the duration of each slide and the animations in PowerPoint to match Christel's narration. Christel mainly paid attention to the content to check if the materials could support the emotions she wanted to express.

Firstly, Christel added a slide at the beginning to acknowledge all of the team's work and the whole group of participants. Secondly, we added a background image (Figure 5) to the part of the story where the potty is passed from child to child. The image comes from a towel where Christel's granddaughter and her school friends drew portraits of themselves providing a personal link to the present. Thirdly, we improved the animation to add twinkling stars and a short animation of music symbols that follow the background music. When we finished the first version of the complete story, Diego suggested adding subtitles, and Christel decided to create a version with English subtitles and one with German subtitles that she translated herself. We made over four versions of the story project based on the changes in the process and it took several iterations to finish all the details. After doing these changes, we presented the story to the group. Christel was delighted with the final result and asked for a way to share the file with her family.

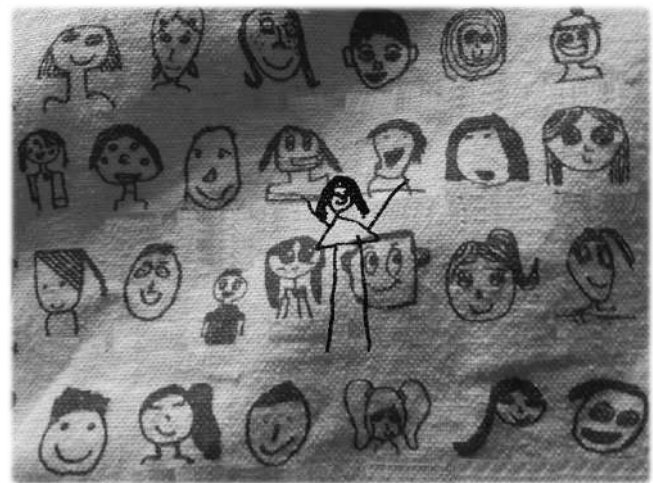


Figure 5: The towel of Christel's granddaughter and her school friends' portraits

5.7 Music: the thread that elevates the story

The most important moment of the story for Christel is the moment when her little brother starts singing “Weisst Du Wieviel Sternlein

Stehen". The emotional connection between this song, her memory and the emotions that Christel wanted to convey in the story raised the importance of working on different audio elements that would represent and highlight the significance of the song. Christel and Diego worked together to combine four audios into one to create the audio of the story. The four parts are Christel's narration, background music, the song and the Martin Luther Homes Choir singing.

5.7.1 Christel's narration. It was important for Christel that she would do the narration herself. After getting to the final version of the plot of the story, Sonja recorded Christel reading the script. For Christel, the narration had to sound natural and spontaneous, so she decided to keep the recording even though her voice was quavering due to her emotions when reading the script. The recording of the narration was only done twice, and we used the second recording in the video.

5.7.2 Background music for the narration. Christel wanted neutral background music that supported her narration until the moment that her brother started signing. Diego brought a short list of potential background songs to use. Christel decided which song to use during a session where they listened to these songs.

5.7.3 Weisst Du Wieviel Sternlein Stehen: the shining star of the story. The climax of the story came when Christel's little brother looked up to the stars, he started singing the song, and then Christel's family and all the other families around joined in. These moments brought many emotions to Christel: she felt peaceful and happy, knowing that they all would be looked after for that night. Then, it was important to find a recording of the song itself that aligned with the context of the story and with the emotions of Christel. Christel and Diego listened to a few versions of the song "Weisst Du Wieviel Sternlein Stehen" during a session, and Diego also looked for other versions after that session. In the following session, Diego and Christel decided to use a version sung by a European Boys' Choir that started with a few children singing the first verses, and then the rest of the Choir joined them. Diego got the authorisation from the Choir representatives to use their version of the song in the story.

Christel and Diego decided to fade out the piano-based background music and use the song when the little brother looks up and starts singing. To align the narration and the song, Diego split the audio recording of the narration so the first verses could be heard without any voice over the music. Then, the narration starts again when the rest of the Choir joins. This coordination between narration and music linked the emotions perceived by Christel when all the families were singing and her own feelings at that moment.

5.7.4 The Martin Luther Homes Choir singing. Christel participates in the local Church community at the Residential Village she lives in. She had talked about the digital story to other residents, and they knew about the importance of this song to her. At that time, the pastor was going to move to another community, and she asked everyone in their Church Choir to sing the song along with the pastor as a farewell to him. Christel translated the lyrics from German to English while maintaining the proper metrics, rhythm and mood of the original song. To record the local Choir by using her smartphone, Christel asked Diego to teach her, and they practised

several times until she was able to do it herself. Then, she emailed the recording to Diego. Christel and Diego decided to include the recording and the translated lyrics at the end of the story to acknowledge the importance of the local community to Christel.

6 LESSONS LEARNED

6.1 The importance of emotional goals

The visual materials and the audio contributed to conveying subtle emotions in Christel's story. It is important to capture and share the revealed emotions in the story, which will enrich the experience of both the storyteller and the audience. In this iterative story development process with Christel, we learned the importance of emotions in her story during the collaboration. For instance, she wanted to convey happiness and safety when they sang. This focus on emotions was also the reason she decided to present the song story instead of the bread story.

that [bread] story has a sad twist because my mother really would have liked that we didn't eat the bread so quickly... The joy that I had when we were singing, that was the reason for my choice...Because the positive [feelings of the song story].

The digital storytelling format supported Christel to articulate her emotions in a new format and include meaningful elements [3]. The opportunities to create digital materials that could augment the meaning of the story created a link from the past into the present.

Because the story of the stars is known in my family, they grew up with it and know about it. And when you were saying try something else... [I realised that] this would be a wonderful opportunity to learn something to do a different [version]...You put that spark into us.

Engaged by building the story differently, Christel kept bringing new ideas and collaborating with us to concretise these ideas. For instance, the drawing for the potty, teddy and backpack, the towel from her granddaughter, the local community Choir and the English translation of the lyrics. All these new elements added a personal touch, impact and meaning to the story.

6.2 Building confidence through positive experience

At the beginning, Christel felt it was difficult to produce these materials based on her ability to use technology. It also challenged us to figure out how to collaborate with her. We attempted to show her some technologies that could be useful for building her story, and we wanted to start in a simple way that would not overwhelm her. When introducing the digital drawing tool, we started with basic use. After she created the landscape and got interested in it, we showed more complex features that would be needed for creating the images of her story. After creating the train image, she explored other features in Sketchbook in her own time and spontaneously looked for strategies when facing difficulties. Now she keeps drawing with Sketchbook in her leisure time and expressed her appreciation to the research team for giving her a new tool to do something meaningful and enjoyable.

To be here with the others and you and to learn something and it is so relaxed, it is so friendly. And I learned

and I think especially the drawing on the computer will enrich my [life]. Not at the moment, I'm too busy. But one day when I'm by myself and have a lot of time, it will again, enrich my life... And I am very thankful for your encouragement.

From Christel's digital drawing experience we see a positive loop of technology learning. Although she started with uncertainty about what she can do, her confidence and interest in digital drawing kept building up from the first successful trial of Sketchbook. This experience aligns with previous works that encourage a perspective on older adults as digital content producers [13]. Encouraged by the first use, she kept learning and practising, and her skills and confidence kept growing during the process [2, 7, 10].

7 CONCLUSION

In this case study, we have presented the development of Christel's story in a co-design storytelling project and provided insights about how to help older adults learn about technology while doing meaningful activities. Through an iterative process, we learned that the co-design approach can help the research team understand the goals of older adults, particularly their emotional goals, to then provide suitable assistance for them to learn to use technology and build their confidence. In the future, we plan to conduct different activities using technologies in the group to learn from their experiences.

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